

FASTT: Feature-Adaptive Speech-to-Text Model Selection

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ABSTRACT

Existing approaches to automatic speech recognition (ASR) either rely on a single speech-to-text (STT) model or combine multiple models through computationally expensive ensemble techniques, leading to suboptimal trade-offs between accuracy and efficiency across diverse acoustic conditions. Since different STT models exhibit complementary strengths, selecting the most suitable model for each input can significantly improve transcription accuracy. Here, we introduce FASTT, a feature-adaptive STT selection framework that learns to select the most appropriate pretrained STT model for each input. FASTT jointly learns a task-aligned feature transformation and a lightweight selection function based on acoustic and embedding-derived features, and is trained in an end-to-end manner to minimize a cumulative transcription-error objective. Due to this end-to-end training, we learn both the features that best distinguish between STT models and the selection algorithm so that FASTT consistently outperforms static selection and standard routing methods. Our experiments over multiple challenging datasets show that FASTT achieves consistent improvements over strong baselines, with up to 13.5% relative WER reduction compared to traditional ensemble methods, and up to 5% relative improvement over classical feature selection approaches. FASTT maintains near real-time inference with millisecond-scale selector overhead, enabling efficient deployment. Furthermore, we provide convergence guarantees showing that the proposed training procedures converge to first-order stationary points under mild conditions, ensuring stable and reliable optimization, and we publicly release the code to support reproducibility.

1. Introduction

Automatic speech recognition (ASR) has advanced rapidly with the emergence of large-scale pretrained models and self-supervised representation learning (Latif, Zaidi, Cuayáhuítl, Shamshad, Shoukat, Usama and Qadir, 2025; Sun, Zhang, Xie and Zhang, 2017; Baevski, Zhou, Mohamed and Auli, 2020; Hsu, Bolte, Tsai, Lakhota, Salakhutdinov and Mohamed, 2021; Radford, Kim, Xu, Brockman, McLeavey and Sutskever, 2023; Mohamed, Lee, Borgholt, Havtorn, Edin, Igel, Kirchhoff, Li, Livescu, Maaløe, Sainath and Watanabe, 2022). Modern speech-to-text (STT) systems such as transformer encoder-decoder architectures and transducer-based models achieve strong performance across diverse domains (Latif et al., 2025; Prabhavalkar, Hori, Sainath, Schlüter and Watanabe, 2024; Dong, Xu and Xu, 2018; Gulati, Qin, Chiu, Parmar, Zhang, Yu, Han, Wang, Zhang, Wu and Pang, 2020). However, despite their scale and robustness, no single STT model consistently achieves optimal accuracy across all acoustic conditions (Sun et al., 2017; Zhu et al., 2024; Zhu, Zhang, Zhang and Dai, 2023; Cornell et al., 2025). In real-world deployments, audio inputs demonstrate variability in noise, channel characteristics, speaker identity, and speaking style, leading to input-dependent performance variation across models (Cornell et al., 2025; Watanabe et al., 2020). As a result, different STT models demonstrate complementary strengths, motivating adaptive model selection (Rice, 1976; Vanschoren, 2019; Vettoruzzo et al., 2024). While ensemble methods such as ROVER reduce errors by aggregating multiple hypotheses (Fiscus, 1997), they incur significant computational overhead. In contrast, adaptive selection aims to exploit model diversity while maintaining near single-model efficiency.

A key challenge in adaptive selection is predicting model performance from input-dependent features (Gamper, Emmanouilidou, Braun and Tashev, 2020; Ragni et al.,

2022; Kastanos, Ragni and Gales, 2020). Modern acoustic representations often use a high number of spectral features, prosodic statistics, signal-quality measures, and high-dimensional neural embeddings to distinguish between STT algorithms based on the input audio (Gamper et al., 2020; Zezario, Fu, Chen, Fuh, Wang and Tsao, 2023; Chen, Wang, Chen, Wu, Liu, Chen, Li, Kanda, Yoshioka, Xiao et al., 2022). While expressive, such feature spaces may contain substantial redundancy, correlated dimensions, and weakly informative components, especially when the amount of labeled data available to train the model selection algorithm is limited. In this regime, directly learning a selector over the raw feature space can lead to overfitting, unstable selection behavior, and poor generalization across acoustic domains. The key insight underlying this work is that selection quality depends not only on the selection architecture, but also on the geometry of the feature space in which selection decisions are made (Zezario et al., 2023; Kansizoglou, Bampis and Gasteratos, 2022). Here, we introduce FASTT, a feature-adaptive selection framework that learns a task-aligned feature transformation together with a selection function under a cumulative transcription-error objective. By explicitly adapting the feature representation to the selection loss and the selection function in an end-to-end manner, our approach significantly improves generalization across heterogeneous acoustic conditions while maintaining computational efficiency at inference time.

FASTT learns to identify speech characteristics that distinguish the performance regimes of different STT models and to exploit these characteristics to select the most suitable transcription model for each input. Through the learnable feature transformation module, the framework emphasizes attributes most informative for discriminating between candidate models. Because only one model is executed at inference time, the computational cost remains comparable to that

of a single-model system. In addition to the algorithmic design, we provide convergence guarantees, showing that the proposed training procedure converges to first-order stationary points under mild conditions.

1.1. Related Work

Prior work has focused on improving ASR robustness under acoustic variability, including noise, channel distortion, reverberation, and speaker diversity (Sun et al., 2017; Qian, Bi, Tan and Yu, 2016; Fan et al., 2021; Ghorbani and Hansen, 2022). Techniques such as multi-condition training, data augmentation, domain adaptation, and large-scale self-supervised pretraining have significantly improved model generalization across diverse environments (Escalona, Campos-Roca and Sánchez, 2025; Chen et al., 2022; Peso Parada, Dobrowolska, Saravanan and Ozay, 2022; Park, Chan, Zhang, Chiu, Zoph, Cubuk and Le, 2019; Paraskevopoulos, Kouzelis, Rouvalis, Katsamanis, Katsouros and Potamianos, 2024). These approaches aim to train a single model that performs robustly across conditions by increasing representational capacity and data coverage (Baevski et al., 2020; Prabhavalkar et al., 2024; Chen et al., 2022; Ghorbani and Hansen, 2022). Rather than relying on a single robust model, traditional ensemble methods aim to reduce transcription error by aggregating hypotheses from multiple ASR models (Mangu, Brill and Stolcke, 2000; Hoffmeister et al., 2006; Goel and Byrne, 2000). ROVER (Fiscus, 1997) aligns hypotheses via word-level alignment and performs majority voting, with extensions including confusion-network alignment, consensus decoding, lattice-based combination techniques, and minimum Bayes risk decoding (Mangu et al., 2000; Hoffmeister et al., 2006; Goel and Byrne, 2000; Goel, Byrne and Khudanpur, 2004). More recent neural approaches have explored posterior averaging, logit-level fusion, and ensemble decoding (Liu et al., 2021; Kano et al., 2022; Gitman, Lavrukhin, Laptev and Ginsburg, 2023). These methods effectively exploit complementary strengths across models and often yield measurable reductions in transcription error. Building on this line of work, we consider a complementary formulation that focuses on input-adaptive expert selection rather than hypothesis fusion. Instead of combining multiple model outputs, our framework selects a single pretrained STT expert for each input based on its acoustic characteristics. This retains model diversity while maintaining near single-model inference cost, since only one expert is executed and the selection decision is computed via a lightweight function over pre-extracted features.

Adaptive expert selection is related to mixture-of-experts (MoE) models (Gan, Ning, Qi and Yu, 2026; Jacobs, Jordan, Nowlan and Hinton, 1991; Shazeer, Mirhoseini, Maziarz, Davis, Le, Hinton and Dean, 2017; Fedus, Zoph and Shazeer, 2022; You, Feng, Su and Yu, 2021), in which a gating network assigns input-dependent weights to multiple expert components. Classical MoE architectures jointly train both the experts and the gating function within a single end-to-end model, typically sharing intermediate representations

and optimizing a unified likelihood objective. More recent sparse and conditional computation frameworks, such as sparsely gated MoE layers and Switch Transformers (Gan et al., 2026; Shazeer et al., 2017; Fedus et al., 2022), dynamically activate subsets of parameters to improve scalability and computational efficiency. However, our setting differs fundamentally from standard MoE formulations in several respects. First, the experts in our framework are independently pretrained, large-scale STT systems whose parameters remain fixed during selector training. There is no shared backbone, parameter sharing, or joint optimization between the experts and the selection mechanism. This design allows the framework to incorporate arbitrary pretrained STT systems without requiring additional joint training or fine-tuning. Second, the selection objective is not maximum likelihood or classification accuracy within a unified model, but cumulative transcription-error minimization across heterogeneous acoustic conditions. This is important because the ultimate goal of the system is to reduce transcription error in practice.

A related line of work considers confidence-based and quality estimation (QE) approaches for ASR output selection and routing (Kastanos et al., 2020; Li, Qiu, Zhang, Li, He, Woodland, Cao and Strohman, 2021). Posterior probability-based confidence scores can guide rescoring, hypothesis selection, or fallback decisions (Kastanos et al., 2020; Mangu et al., 2000; Ravi, Raj and Arora, 2024; Li et al., 2021). Domain classification and environment detection have also been explored for routing utterances to domain-specialized models (Gitman et al., 2023; You, Feng, Su and Yu, 2022; Kwon and Chung, 2023). However, confidence-based approaches require at least partial decoder execution to compute posterior scores, while domain classifiers only capture coarse categorical distinctions (Kastanos et al., 2020; Mangu et al., 2000; Gitman et al., 2023; Li et al., 2021). In contrast, FASTT operates entirely on pre-extracted acoustic features without executing any STT decoder at selection time, and learns fine-grained, continuous selection distributions that capture subtle performance differences across experts.

1.2. Contributions

Our contributions are summarized as follows:

- We introduce FASTT, a general-purpose framework for adaptive STT model selection that jointly learns a task-aligned feature transformation and a selection network under a cumulative transcription-error objective. The framework selects the most suitable STT model for each audio input with minimal computational overhead beyond a single model execution.
- We present a principled approach for feature adaptation within the framework, including diagonal gating, linear, low-rank, and nonlinear transformation variants with sparsity and stability regularization. These adaptations improve selection reliability in high-dimensional and data-limited settings.

- We provide two optimization paths for joint training of the feature transformation and the selector: (i) end-to-end backpropagation for differentiable selectors, and (ii) alternating optimization for non-differentiable selectors such as gradient-boosted trees.
- The framework is loss-agnostic, feature-agnostic, and selector-agnostic: any transcription quality metric such as word error rate (WER) or character error rate (CER) can be used (Li, Deng, Gong and Haeb-Umbach, 2014), any discriminative audio feature set can be used, and any learning algorithm can serve as the selection function.
- We provide our implementation as open-source code to support reproducibility and enable researchers to extend the framework with custom features, selectors, and expert models.¹

1.3. Organization

The remainder of the paper is organized as follows. Section 2 formalizes the supervised expert-selection problem. Section 3 presents the proposed feature-adaptive selection framework, including the learnable transformation and the selection function. Section 4 presents convergence guarantees to first-order stationary points for the proposed selectors. Section 5 describes the experimental setup and empirical evaluation. Finally, Section 6 concludes the paper.

2. Problem Formulation

We consider the problem of adaptive STT model selection for a collection of audio inputs $\{s_i\}_{i=1}^N$, where N may be large and the acoustic characteristics may vary substantially across inputs. Each input can demonstrate diverse properties, including noise type and level, channel conditions, speaker identity, accent, speaking rate, and articulation variability, which can significantly affect the transcription accuracy of different STT systems (Yu and Deng, 2015; Li et al., 2014).

Let $\mathcal{E} = \{E_1, E_2, \dots, E_L\}$ denote a pool of L pretrained STT experts. For an input utterance s_i and expert E_j , the produced transcription is written as $y_{i,j} = E_j(s_i)$. Given the reference transcription y_i^{ref} , the quality of the produced transcript is measured using an error function

$$\ell(y_{i,j}, y_i^{\text{ref}}) \in [0, 1],$$

which may correspond to WER, CER, or another normalized transcription metric (Li et al., 2014).

In practice, no single STT model is uniformly optimal across heterogeneous acoustic conditions (Yu and Deng, 2015; Radford et al., 2023; Li et al., 2014). Instead, the optimal model achieving the lowest transcription error may vary from one utterance to another depending on the underlying speech characteristics of the input s_i . This motivates the formulation of an expert selection problem in which the system must decide which expert should process each

input utterance. Let each input utterance s_i be represented by a feature vector $\mathbf{z}_i \in \mathbb{R}^p$ computed using a fixed feature-extraction pipeline. A routing policy is defined as a function $\pi : \mathbb{R}^p \rightarrow \{1, \dots, L\}$, which assigns an expert index to each feature vector. The transcription loss incurred by a routing policy π over the dataset is defined as

$$J(\pi) = \frac{1}{N} \sum_{i=1}^N \ell(y_{i,\pi(\mathbf{z}_i)}, y_i^{\text{ref}}).$$

The objective of adaptive expert selection is to identify the optimal routing policy within a class of admissible policies Π :

$$\pi^\dagger = \arg \min_{\pi \in \Pi} J(\pi).$$

The main challenge in solving this problem comes from the potentially high dimensionality of the feature representation and the limited availability of labeled training examples for learning the routing policy. Furthermore, the relative performance of different STT experts may depend on subtle variations in the acoustic properties of the speech signal. We next introduce a learning-based selection framework that computes an input-dependent distribution over the available STT experts using features extracted from the input audio. To improve selection reliability in high-dimensional feature spaces with limited labeled data, the framework jointly learns a task-aligned feature transformation and a selection network that predicts expert suitability. The algorithm selects a single STT expert from this computed probability distribution to obtain the final transcription in an effective manner.

3. FASTT: Feature-Adaptive Speech-to-Text Model Selection

3.1. Algorithm Overview

The algorithm consists of two phases: an offline training phase and an online inference phase, as illustrated in Fig. 1. In the offline phase, the system learns how to predict the relative suitability of different STT experts based on input-dependent audio characteristics. For each labeled training input s_i , the framework extracts a feature vector $\mathbf{z}_i$ using the feature pipeline described in Section 3.2. This feature representation includes acoustic statistics, prosodic descriptors, signal-level measures, and neural speech embeddings aggregated at the utterance level. To improve selection reliability in high-dimensional feature spaces, the framework learns a parametric feature transformation that maps the original feature vector $\mathbf{z}_i$ into a task-aligned representation, as shown in Fig. 1. This transformation reshapes the feature space to better capture the relationship between acoustic characteristics and expert performance. The transformed features are then passed to a selection function that produces a weight vector over the candidate STT experts $\{E_1, \dots, E_L\}$. The parameters of the feature transformation and the selection function are learned during the offline phase using evaluations of all experts on labeled audio clips, minimizing the cumulative transcription loss defined in Section 2.

¹Code will be released upon publication.

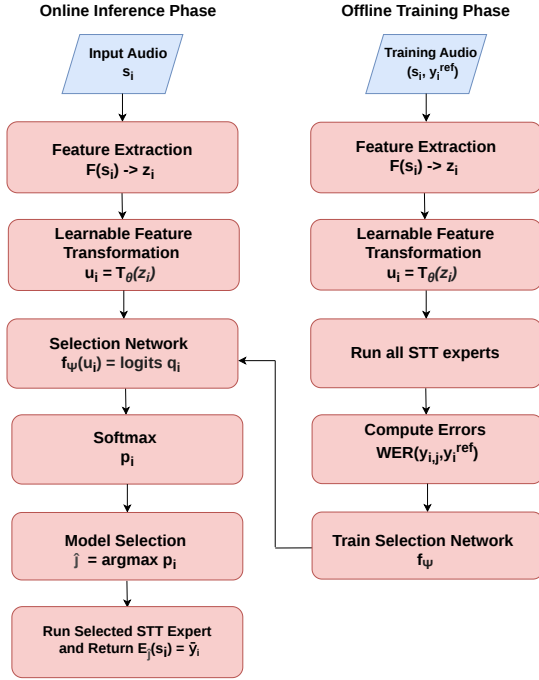


Figure 1: Overview of the FASTT framework. In the offline phase, acoustic features are extracted, transformed through a learnable feature adaptation module, and used to train the selection function by minimizing an expected transcription-error objective. At inference time, only the selected STT expert is executed, introducing minimal overhead beyond a single model run.

In the online inference phase, given a new input audio s_i , the system first extracts the feature vector z_i , as shown in Fig. 1. The trained selection model then computes a weight vector $\mathbf{p}_i$ over the candidate experts based on the transformed feature representation. A single expert E_{j_i} is selected according to the highest predicted probability, and the final transcription $\hat{y}_i = E_{j_i}(s_i)$ is obtained by executing only that expert. Because only one STT model is executed at inference time, the framework introduces only minimal computational overhead beyond a single model run while still exploiting performance diversity across multiple pretrained experts. The complete online inference procedure is summarized in algorithm 1, and the full pipeline is illustrated in Fig. 1.

3.2. Feature Representation

For each incoming audio input s_i , we construct a compact feature vector $z_i \in \mathbb{R}^p$ to encode acoustic and contextual information relevant to STT performance, as shown in Fig. 1. These features summarize signal quality, phonetic content, speaker characteristics, and temporal dynamics (Rabiner and Juang, 1993; Li, Zhao, Klejch, Bell and Lai, 2023).

3.2.1. Spectral-Phonetic Features

We extract 13 Mel-frequency cepstral coefficients (MFCCs) together with their first- and second-order derivatives (Δ and

Algorithm 1 Online STT model selection via FASTT

Inputs:

- Unlabeled input audio clip s_i
- Set of audio feature extraction functions $\mathbb{F}(\cdot)$
- Trained FASTT selection pipeline f_{FASTT}
- STT expert pool $\mathcal{E} = \{E_1, \dots, E_L\}$

Output:

- Final transcription $\hat{y}_i$ for audio s_i

Inference Steps:

- 1: Extract utterance-level features:

$$z_i \leftarrow \mathbb{F}(s_i)$$

- 2: Apply the trained FASTT pipeline to compute selection logits:

$$\mathbf{q}_i \leftarrow f_{\text{FASTT}}(z_i)$$

- 3: Normalize $\mathbf{q}_i$ via softmax to obtain expert probabilities $\mathbf{p}_i$

- 4: Select the expert index:

$$\hat{j}_i \leftarrow \arg \max_{j \in \{1, \dots, L\}} p_{i,j}$$

- 5: Return final transcription:

$$\hat{y}_i \leftarrow E_{\hat{j}_i}(s_i)$$

$\Delta\Delta$) (Davis and Mermelstein, 1990; Furui, 1986). Lower-order coefficients represent the coarse spectral envelope, while higher-order terms encode finer phonetic detail. The inclusion of temporal derivatives provides robustness to coarticulation, prosodic variation, and speaker-specific articulation (Rabiner and Juang, 1993).

3.2.2. Prosodic Features

Prosodic features, including pitch statistics (mean, range) and silence ratios derived from voice activity detection (VAD), characterize speaking rhythm, stress, and expressiveness (Sohn, Kim and Sung, 1999). Such features are particularly valuable for identifying conditions where transcription accuracy may degrade under expressive, emotional, or noisy speech (Li et al., 2023).

3.2.3. Neural Speech Embeddings

High-level speech representations are extracted from pretrained self-supervised encoders such as Wav2Vec2 and HuBERT (Baevski et al., 2020; Hsu et al., 2021). These embeddings encapsulate rich information about phonetic identity, speaker traits, and semantic context, complementing conventional low-level features. Because they require only a single encoder forward pass without full decoding, they introduce negligible overhead relative to STT inference itself. Specifically, the SSL encoder forward pass typically requires less than 50 ms per utterance on modern GPU hardware, which is an order of magnitude smaller than full ASR decoding time. For resource-constrained deployment scenarios, lighter embedding models or knowledge-distilled variants could be substituted without modifying the framework.

3.2.4. Signal and Temporal Features

Signal-level features including signal-to-noise ratio (SNR), zero-crossing rate (ZCR), and spectral entropy quantify acoustic clarity and spectral irregularity (Schuller and Batliner, 2013; Eyben, Scherer, Schuller, Sundberg, André, Busso, Devillers, Epps, Laukka, Narayanan and Truong, 2016). Temporal features such as utterance duration, speech rate, and pause ratio measure fluency and pacing, both of which significantly impact STT robustness (Ko, Peddinti, Povey and Khudanpur, 2015; Tripathi, Kim, Zhang, Lu and Sak, 2020). The resulting feature vector $\mathbf{z}_i$ thus integrates spectral, prosodic, neural, and temporal features into a unified contextual representation for each input. This vector serves as the conditioning signal for the supervised selection function.

Remark 1. *The proposed framework is feature-agnostic. The selection network operates on a generic feature vector and can incorporate any set of discriminative features, such as acoustic, linguistic, prosodic, model-intrinsic, or domain-specific features that correlate with relative STT performance. If alternative features provide better separation between candidate algorithms for a given application, they can be substituted or augmented without modifying the training procedure, loss function, or selection mechanism.*

3.3. Learnable Feature Transformation

The feature representation $\mathbf{z}_i \in \mathbb{R}^p$ described in Section 3.2 captures diverse acoustic and embedding-based properties of the input audio. While such descriptors provide informative signals for expert selection, their raw representation may not be optimally structured for predicting relative expert performance. In particular, high-dimensional feature spaces may contain correlated dimensions or redundant information that can make the selection model more difficult to train when labeled data are limited.

Classical approaches to dimensionality reduction in such settings include filter-based methods, such as mutual information ranking and variance thresholding, wrapper-based strategies, such as sequential feature selection (Guyon and Elisseeff, 2003; Kohavi and John, 1997), and embedded methods that couple feature selection with model training (Friedman, 2001). However, filter methods disregard inter-feature dependencies and their interaction with the downstream objective, wrapper approaches have combinatorial search costs that scale poorly with p , and embedded methods are typically tied to specific model families. These limitations motivate a learnable, task-aligned feature transformation that can be jointly optimized with any downstream selector.

We introduce a parametric mapping applied prior to the selection network that reshapes the geometry of the feature space without modifying the feature extraction pipeline itself, as illustrated in Fig. 1. The transformed representation is defined as

$$\mathbf{u}_i = \mathbb{T}_\theta(\mathbf{z}_i),$$

where $\mathbb{T}_\theta : \mathbb{R}^p \rightarrow \mathbb{R}^{d'}$ denotes a parametric transformation with learnable parameters θ , and $\mathbf{u}_i$ serves as the input to

the downstream expert selection function. The optimization strategy for θ depends on whether the downstream selector is differentiable, as discussed in Section 3.5. We present several transformation variants below; all are compatible with any selector variant within the framework.

3.3.1. Diagonal Feature Gating

As a capacity-controlled transformation, we consider feature-wise scaling:

$$\mathbf{u}_i = \mathbf{g} \odot \mathbf{z}_i,$$

where $\mathbf{g} \in \mathbb{R}^p$ is a learnable gating vector and $\odot$ denotes element-wise multiplication. Each component g_j assigns an importance weight to the corresponding feature dimension j . This formulation acts as a differentiable feature-selection mechanism: dimensions with small g_j values are effectively suppressed, while informative features receive larger weights. Because the number of additional parameters is limited to p , the diagonal gating transformation provides a simple yet effective mechanism for controlling overfitting in high-dimensional, data-limited regimes. To promote sparsity and stability, we apply combined ℓ_1/ℓ_2 regularization:

$$\lambda_1 \|\mathbf{g}\|_1 + \lambda_2 \|\mathbf{g}\|_2^2.$$

3.3.2. Linear Feature Transformation

A more expressive alternative is a full linear projection:

$$\mathbf{u}_i = \mathbf{W}\mathbf{z}_i + \mathbf{b},$$

where $\mathbf{W} \in \mathbb{R}^{d' \times p}$ is a learnable projection matrix and $\mathbf{b} \in \mathbb{R}^{d'}$ is a bias vector. This transformation allows the model to reweight and mix the original feature dimensions in a task-aligned manner. By choosing $d' < p$, the transformation acts as a dimensionality-reduction mechanism. Weight decay regularization is applied to $\mathbf{W}$ to control overfitting.

3.3.3. Low-Rank Linear Transformation

While the full linear transformation allows arbitrary mixing of feature dimensions, it may introduce a large number of parameters when p is high. To provide a more capacity-controlled alternative, we consider a low-rank factorization of the projection matrix:

$$\mathbf{u}_i = \mathbf{W}_2 \mathbf{W}_1 \mathbf{z}_i,$$

where $\mathbf{W}_1 \in \mathbb{R}^{r \times p}$, $\mathbf{W}_2 \in \mathbb{R}^{d' \times r}$, and $r \ll \min(p, d')$ is a bottleneck dimension. This factorization constrains the rank of the projection, allowing the model to capture interactions between feature dimensions while reducing the number of learnable parameters.

3.3.4. Nonlinear Feature Transformation

To capture nonlinear feature interactions that may affect relative expert performance, we consider a nonlinear bottleneck transformation:

$$\mathbf{u}_i = \mathbf{W}_2 \sigma(\mathbf{W}_1 \mathbf{z}_i),$$

where $\sigma(\cdot)$ is an element-wise activation function such as ReLU or GELU, $\mathbf{W}_1 \in \mathbb{R}^{r \times p}$ and $\mathbf{W}_2 \in \mathbb{R}^{d' \times r}$ are learnable weight matrices, and $r \ll \min(p, d')$ is the bottleneck dimension. The bottleneck structure constrains the parameter count and helps prevent overfitting when labeled data are limited.

Remark 2. *The proposed framework is flexible with respect to the choice of feature transformation. In addition to the variants presented here, other parametric transformations such as deeper neural encoders, kernel-based mappings, or representation learning modules could be used without modifying the overall pipeline. All transformation variants are evaluated in Section 5.*

3.4. Selection Objective

Given the transformed feature vector $\mathbf{u}_i \in \mathbb{R}^{d'}$ obtained from the feature transformation stage described in Section 3.3, the goal of the selection function is to assign preference scores to a set of L pretrained STT experts $\mathcal{E} = \{E_1, \dots, E_L\}$. We define the selection function as a parametric mapping

$$f_\psi(\mathbf{u}_i) = \mathbf{q}_i \in \mathbb{R}^L,$$

where ψ denotes the trainable parameters and $\mathbf{q}_i = (q_{i,1}, \dots, q_{i,L})$ represents the logits associated with the candidate STT experts. Each component $q_{i,j}$ reflects the relative suitability of expert E_j for the audio input $\mathbf{s}_i$. We apply a softmax operation to the logits to obtain a normalized selection probability for each expert:

$$p_{i,j} = \frac{\exp(q_{i,j})}{\sum_{m=1}^L \exp(q_{i,m})}, \quad j = 1, \dots, L, \quad (1)$$

where $p_{i,j}$ denotes the probability assigned to expert E_j . Given these probabilities, we define the expected transcription loss for the audio clip $\mathbf{s}_i$ as

$$\mathcal{L}_i = \sum_{j=1}^L p_{i,j} \text{WER}(y_{i,j}, y_i^{\text{ref}}), \quad (2)$$

where y_i^{ref} is the reference transcription. The WER between a hypothesis transcription $y_{i,j}$ and the reference y_i^{ref} is defined as

$$\text{WER}(y_{i,j}, y_i^{\text{ref}}) = \frac{S + I + D}{|y_i^{\text{ref}}|},$$

where S , I , and D denote the numbers of substitutions, insertions, and deletions obtained from a standard Levenshtein alignment (Li et al., 2014).

Remark 3. *Although we use WER as the optimization objective due to its widespread use and interpretability in speech recognition research (Yu and Deng, 2015; Li et al., 2014), other transcription quality metrics such as CER or task-specific error measures could also be used within the same framework.*

The overall training objective minimizes the expected transcription error across the dataset:

$$\min_{\theta, \psi} \mathcal{L} = \frac{1}{N} \sum_{i=1}^N \mathcal{L}_i = \frac{1}{N} \sum_{i=1}^N \sum_{j=1}^L p_{i,j} \text{WER}(y_{i,j}, y_i^{\text{ref}}). \quad (3)$$

During training, the WER values $\text{WER}(y_{i,j}, y_i^{\text{ref}})$ are treated as fixed constants obtained from offline evaluation of the STT experts. The optimization strategy for the feature transformation parameters θ and the selection function parameters ψ depends on whether the selection function is differentiable with respect to its inputs and parameters, as discussed in the following subsection.

3.5. Joint Feature and Model Selection

A key contribution of the FASTT framework is the joint training of the feature transformation $\mathbb{T}_\theta$ and the selection function f_ψ . Instead of performing feature selection and model selection as two separate steps, FASTT learns both components together using the expected transcription error objective. This joint training aligns the feature representation directly with the model selection task.

The framework supports two optimization paths depending on the differentiability of the selection function.

3.5.1. FASTT for Differentiable Model Selectors

When the selection function $f_\psi(\cdot)$ is differentiable with respect to its parameters ψ and its inputs $\mathbf{u}_i$, the parameters θ and ψ are optimized jointly using standard backpropagation. Gradients of the expected word error rate (WER) objective flow from the softmax probabilities through the selection network and into the feature transformation. This strategy enables end-to-end feature adaptation in a single training loop without requiring auxiliary optimization steps.

As a representative differentiable model, we use Soft Decision Trees (SDTs). SDTs are hierarchical models in which each internal node makes a soft, probabilistic routing decision based on a linear combination of features. Because these routing decisions are continuous rather than fully discrete, SDTs are differentiable and suitable for gradient-based training. We construct our selector, FASTT-SDT, using an additive ensemble of SDTs, as in Karaca and Kozat (2026). The model consists of B sequential boosting rounds. At each round b , a per-round feature transformation $\mathbb{T}_{\theta_b}(\cdot)$ is applied to the input features, followed by a soft decision tree $h_b(\cdot)$ producing L logits. The final output is the sum of logits across all boosting rounds:

$$\mathbf{q}_i = \sum_{b=1}^B h_b(\mathbb{T}_{\theta_b}(\mathbf{z}_i)). \quad (4)$$

At each round, the current tree is fit on the gradient of the previous predictions. The per-round feature transformation $\mathbb{T}_{\theta_b}$ and the soft decision tree h_b are optimized jointly using backpropagation on the expected-WER objective in eq. (3). A key property of the differentiable FASTT selector is that each boosting round learns its own feature transformation,

Algorithm 2 Offline training: FASTT for differentiable selectors

Inputs:

- $\{(\mathbf{s}_1, y_1^{\text{ref}}), \dots, (\mathbf{s}_N, y_N^{\text{ref}})\}$: labeled audio clips with reference transcriptions
- $\mathbb{F}$: feature extraction pipeline
- $\mathcal{E} = \{E_1, \dots, E_L\}$: pretrained STT expert pool
- B : number of boosting rounds

Output: Trained FASTT model $\{(\mathbb{T}_{\theta_b}, h_b)\}_{b=1}^B$

- 1: **Offline data preparation:**
- 2: **for** $i \leftarrow 1$ **to** N **do**
- 3: Extract features: $\mathbf{z}_i \leftarrow \mathbb{F}(\mathbf{s}_i)$
- 4: **for** each expert $E_j \in \mathcal{E}$ **do**
- 5: Transcribe: $y_{i,j} \leftarrow E_j(\mathbf{s}_i)$
- 6: Compute: $\text{WER}_{i,j} \leftarrow \text{WER}(y_{i,j}, y_i^{\text{ref}})$
- 7: **end for**
- 8: **end for**
- 9: Initialize ensemble prediction: $\hat{\mathbf{q}}_i^{(0)} \leftarrow \mathbf{0}$ for all i
- 10: **Boosted training:**
- 11: **for** round $b \leftarrow 1$ **to** B **do**
- 12: Compute residual targets from $\hat{\mathbf{q}}_i^{(b-1)}$ using eq. (2)
- 13: Initialize per-round feature transform $\mathbb{T}_{\theta_b}$
- 14: Fit base model h_b on $\{(\mathbb{T}_{\theta_b}(\mathbf{z}_i), \text{residual}_i)\}$
- 15: Jointly optimize θ_b and h_b via backpropagation with ℓ_1 regularization on θ_b
- 16: Update ensemble:

$$\hat{\mathbf{q}}_i^{(b)} \leftarrow \hat{\mathbf{q}}_i^{(b-1)} + h_b(\mathbb{T}_{\theta_b}(\mathbf{z}_i))$$

17: **end for**

18: **Online inference:** For a new audio input $\mathbf{s}_i$, compute

$$\mathbf{q}_i = \sum_{b=1}^B h_b(\mathbb{T}_{\theta_b}(\mathbb{F}(\mathbf{s}_i))),$$

apply softmax, and select $\hat{j}_i = \arg \max_j p_{i,j}$

allowing different rounds to focus on different subsets or combinations of the input features. The complete offline training procedure is given in algorithm 2.

3.5.2. FASTT for Non-Differentiable Model Selectors

Tree-based ensemble methods such as gradient-boosted decision trees perform well on structured tabular data and provide strong regularization. However, they are not differentiable with respect to their input features. Gradients cannot be propagated through the discrete tree structure to update the parameters of the feature transformation $\mathbb{T}_{\theta}$. This creates an optimization challenge, since θ and ψ cannot be trained jointly via standard backpropagation.

To enable joint adaptation without requiring differentiability, FASTT uses an alternating optimization strategy (Bertsekas, 2016). We decompose the joint training problem into two subproblems that are optimized iteratively:

1. *Selector update (fix θ , optimize ψ):* Given the current feature transformation $\mathbb{T}_{\theta}$, compute the transformed features $\mathbf{u}_i = \mathbb{T}_{\theta}(\mathbf{z}_i)$ and train the tree-based selector on these features.

2. *Transform update (fix ψ , optimize θ):* With the tree structure fixed, update the feature transformation parameters θ .

This alternating procedure is repeated for a fixed number of iterations. The complete training procedure is given in algorithm 3.

As a representative non-differentiable selector in this path, we utilize XGBoost (Chen and Guestrin, 2016). Crucially, the selector must minimize the expected-WER objective instead of conventional classification or regression losses. Standard objectives such as cross-entropy align with categorical targets, whereas FASTT requires minimizing a continuous, sequence-level transcription error over a probability distribution across candidate STT experts. Therefore, XGBoost is trained with a custom objective function. We supply the algorithm with specifically formulated first- and second-order derivatives to integrate expected-WER minimization into the tree-building process.

Let $g_{i,j}$ and $h_{i,j}$ denote the first- and second-order derivatives of $\mathcal{L}_i$ with respect to the output logit $q_{i,j}$:

$$g_{i,j} = \frac{\partial \mathcal{L}_i}{\partial q_{i,j}} = p_{i,j} (\text{WER}(y_{i,j}, y_i^{\text{ref}}) - \mathcal{L}_i). \quad (5)$$

To reduce computational cost and maintain stable split gains, we use a diagonal approximation to the Hessian:

$$h_{i,j} = \frac{\partial^2 \mathcal{L}_i}{\partial q_{i,j}^2} = p_{i,j}(1 - 2p_{i,j}) (\text{WER}(y_{i,j}, y_i^{\text{ref}}) - \mathcal{L}_i). \quad (6)$$

We clip negative Hessian values to a small positive constant to ensure stable tree building (Chen and Guestrin, 2016).

Remark 4. *The framework is flexible and can accommodate a wide range of learning algorithms for the selection function. In addition to the variants presented here, other architectures such as transformers (Vaswani, Shazeer, Parmar, Uszkoreit, Jones, Gomez, Kaiser and Polosukhin, 2017) or alternative supervised learning algorithms could be used without modifying the overall pipeline.*

3.6. Sampling Strategy

Given the selection probabilities $\mathbf{p}_i = (p_{i,1}, \dots, p_{i,L})$ produced by the selection function, we choose the final expert as

$$\hat{j}_i = \arg \max_{j \in \{1, \dots, L\}} p_{i,j}.$$

The selected expert is then executed to produce the final transcription

$$\hat{y}_i = E_{\hat{j}_i}(\mathbf{s}_i),$$

as shown in Fig. 1. Because only a single STT expert is executed, the computational cost of inference remains comparable to that of a single-model system. The selection function itself introduces negligible overhead since it operates on pre-extracted features and involves only lightweight computation relative to the STT models.

Algorithm 3 Offline training: FASTT for non-differentiable selectors

Inputs:

- $\{(\mathbf{s}_1, y_1^{\text{ref}}), \dots, (\mathbf{s}_N, y_N^{\text{ref}})\}$: labeled audio clips with reference transcriptions
- $\mathbb{F}$: feature extraction pipeline
- $\mathcal{E} = \{E_1, \dots, E_L\}$: pretrained STT expert pool
- $\mathbb{T}_\theta$: learnable feature transformation
- T : number of alternating iterations

Output: Trained feature transformation $\mathbb{T}_\theta$ and selector f_ψ

- 1: **Offline data preparation:**
- 2: **for** $i \leftarrow 1$ **to** N **do**
- 3: Extract features: $\mathbf{z}_i \leftarrow \mathbb{F}(\mathbf{s}_i)$
- 4: **for** each expert $E_j \in \mathcal{E}$ **do**
- 5: Transcribe: $y_{i,j} \leftarrow E_j(\mathbf{s}_i)$
- 6: Compute: $\text{WER}_{i,j} \leftarrow \text{WER}(y_{i,j}, y_i^{\text{ref}})$
- 7: **end for**
- 8: **end for**
- 9: **Alternating optimization:**
- 10: **for** iteration $t = 1, \dots, T$ **do**
- 11: Transform features: $\mathbf{u}_i \leftarrow \mathbb{T}_\theta(\mathbf{z}_i)$ for all i
- 12: Compute softmax weights $\mathbf{p}_i$ via eq. (1)
- 13: Compute per-instance gradients $g_{i,j}$ and Hessians $h_{i,j}$ via eqs. (5) and (6)
- 14: **Selector update:** Train XGBoost f_ψ on $\{\mathbf{u}_i\}$ using the custom WER-based objective
- 15: Obtain updated logits: $\mathbf{q}_i \leftarrow f_\psi(\mathbf{u}_i)$
- 16: **Transform update:** Fix f_ψ and optimize θ using a valid surrogate gradient or continuous approximation
- 17: **end for**
- 18: **Online inference:** For a new audio input $\mathbf{s}_i$, compute

$$\mathbf{q}_i = f_\psi(\mathbb{T}_\theta(\mathbb{F}(\mathbf{s}_i))),$$

apply softmax, and select $\hat{j}_i = \arg \max_j p_{i,j}$

values, Lipschitz-continuous gradients, and a weak-learner edge condition, we obtain a monotone descent property. Let $L(\mathbf{q}^{(n)})$ denote the training loss evaluated at the current logits $\mathbf{q}^{(n)}$.

Under these assumptions, and for a sufficiently small step size η , the training dynamics admit the following descent inequality:

$$L_{n+1} \leq L_n - \gamma \|\nabla L(\mathbf{q}^{(n)})\|^2, \quad (7)$$

where $\gamma > 0$ is a constant that depends on the learning rate, the weak-learner edge, and the smoothness of the loss function. Summing over iterations yields an $\mathcal{O}(1/N)$ convergence rate to a first-order stationary point. Full derivations and the explicit form of γ are provided in Appendix A.

4.2. Neural Selector (MLP / FASTT-SDT)

For the neural selector, we rely on standard non-convex optimization results under smoothness of the loss function and bounded variance of the stochastic gradients. Under these assumptions, the training dynamics satisfy an $\mathcal{O}(1/N)$ convergence rate for full-batch gradient descent and an $\mathcal{O}(1/\sqrt{N})$ convergence rate for mini-batch SGD, implying convergence to a first-order stationary point. Detailed derivations are provided in Appendix A.

Remark 5. *These results characterize the convergence behavior of the selector optimization under fixed or jointly updated feature transformations. Formal convergence guarantees for the end-to-end joint system, including both the feature transformation and the selector, in the non-differentiable alternating-optimization case remain an open question and are left for future investigation.*

4. Mathematical Proof of Convergence

We provide convergence guarantees for the selector optimization procedures presented in Section 3.5. Specifically, we show that, under mild regularity conditions, both training procedures converge to a first-order stationary point of the training objective.

The proofs hold for any smooth loss with bounded curvature, consistent with the framework’s loss-agnostic design. Our convergence analysis focuses on the optimization of the selector parameters for a fixed feature transformation. In the full framework, the transformation parameters are updated either jointly, for differentiable selectors, or through alternating optimization, for tree-based selectors, which preserves the stationary-point guarantees for the selector subproblem. The same convergence guarantees apply to alternative admissible objectives, such as cross-entropy, mean squared error, or other smooth sequence-level surrogates, that satisfy the smoothness assumptions used in the analysis.

4.1. XGBoost-based Selector

For the XGBoost-based selector, we prove convergence under our custom WER-based objective by extending standard second-order boosting theory. Assuming bounded loss

5. Experiments and Results

We evaluate the proposed FASTT framework using the selector and transformation variants described in Sections 3.3 and 3.5. The experiments are designed to (i) compare the framework variants against external baselines and classical feature selection strategies, (ii) ablate the contribution of different feature transformation and selector choices, and (iii) analyze computational overhead.

5.1. Datasets and Feature Extraction

We construct a heterogeneous evaluation corpus by aggregating clips from three publicly available English speech datasets that span a wide range of acoustic conditions. The corpus includes LibriSpeech (Panayotov, Chen, Povey and Khudanpur, 2015) (read audiobook speech from the *test-clean* and *test-other* splits), VoxPopuli’s English accented subset (Wang, Riviere, Lee, Wu, Talnikar, Haziza, Williamson, Pino and Dupoux, 2021) (European Parliament speech with accented English and varying room acoustics), and the challenging SDM subset of AMI (Mccowan, Carletta, Kraaij, Ashby, Bourban, Flynn, Guillemot, Hain, Kadlec, Karaiskos, Kronenthal, Lathoud, Lincoln, Lisowska Masson, Post, Reidsma and Wellner, 2005) (meeting recordings using

Table 1
Dataset statistics.

Dataset	Ami	Libri	VoxP.
Total Clips	29411	16143	4324
Features	1672	1672	1672
Train Size	25000	13721	3675
Test Size	4411	2422	649

single distant microphones). This corpus is designed so that no single STT model dominates across all conditions, providing a meaningful testbed for adaptive model selection. The dataset statistics are summarized in Table 1.

Each clip is represented by a $p = 1672$ -dimensional feature vector $\mathbf{z}_i$ composed of the feature groups described in Section 3.2. Specifically, the vector concatenates 78 spectral-phonetic features (MFCCs with $\Delta/\Delta\Delta$), 17 prosodic and VAD features, 14 channel and noise features, 8 signal complexity features, 1536 neural embeddings (mean-pooled Wav2Vec2 and HuBERT embeddings, 768 dimensions each), and 19 embedding statistics and metadata features. This high-dimensional representation motivates the learnable feature transformation: directly training a selector on all 1672 features in data-scarce regimes can lead to overfitting and unstable selection.

5.2. Models and Baselines

5.2.1. Individual STT Experts

We employ $L = 3$ pretrained STT models as the expert pool $\mathcal{E}$: Whisper-large-v3 (Radford et al., 2023), a transformer encoder-decoder trained on 680k hours of weakly supervised web audio; Parakeet-TDT-0.6b-v2 (NVIDIA, 2025b), a FastConformer-Transducer model optimized for English ASR with token-and-duration transducer decoding; and Canary-Qwen-2.5b (NVIDIA, 2025a), a multi-task encoder-decoder combining a FastConformer audio encoder with a Qwen-2.5 language-model backbone. For each audio clip $\mathbf{s}_i$, all three models are run offline to obtain transcriptions $\{y_{i,j}\}_{j=1}^3$ and their associated word error rates against the reference transcripts.

5.2.2. External Baselines

We compare against non-learned and simple baselines that do not use the FASTT feature transformation pipeline:

- ROVER (Fiscus, 1997), which aligns hypotheses from all L models via pivot-based word alignment and applies majority voting.
- Random Ensemble, which selects a model uniformly at random for each utterance.

5.2.3. Classical Feature Selection and Raw Baselines

To demonstrate the advantage of learned, task-aligned feature transformations within the FASTT framework, we evaluate four baselines representing traditional approaches. These methods either use the full feature set directly or apply static feature selection as a fixed preprocessing step:

- Raw-SDT / Raw-XGBoost: These baselines utilize the complete, high-dimensional feature set without any prior dimensionality reduction or selection. This allows us to evaluate the performance of the selector models, Soft Decision Trees and XGBoost, in their vanilla form and serves as a reference point for whether feature selection or transformation provides empirical benefit.
- SelectKBest-SDT / SelectKBest-XGBoost: To assess the impact of standard dimensionality reduction, we apply a univariate statistical filter, namely ANOVA F-value ranking, to select the top k most informative features as a preprocessing step. The selector models are then trained exclusively on this reduced subset.

All baseline methods are optimized using the same expected-WER objective function. This ensures a fair comparison by isolating the impact of feature processing, contrasting raw input and static feature selection against the dynamic, task-aligned transformations introduced in the FASTT variants.

5.2.4. FASTT Variants

We evaluate two representative variants within the FASTT framework, corresponding to the differentiable and non-differentiable optimization paths described in Section 3.5:

- FASTT-SDT (differentiable path): A boosted feature-adaptive selector with soft decision trees as the base model and diagonal feature gating, trained end-to-end with per-round feature adaptation and ℓ_1 -penalized routing.
- FASTT-XGBoost (non-differentiable path): An XGBoost-based selector with learned diagonal feature gating, trained via the alternating optimization procedure described in Section 3.5.2.

The choice of base model and selector within each path is further studied in the ablation experiments.

5.3. Training Setup and Evaluation

The dataset is split into 80% training and 20% test sets, stratified by source dataset where possible. Within the training portion, 15% is further reserved for validation to facilitate early stopping and hyperparameter selection. To ensure a fair comparison, all learned methods receive an equal computational budget for hyperparameter tuning. We employ Optuna (Akiba, Sano, Yanase, Ohta and Koyama, 2019) with a Tree-structured Parzen Estimator (TPE) sampler. The hyperparameter search spaces are given in Table 2. The primary evaluation metric is the mean WER of the selected model across all test clips:

$$\text{WER}_{\text{method}} = \frac{1}{|\mathcal{D}_{\text{test}}|} \sum_{i \in \mathcal{D}_{\text{test}}} \ell(y_{i,j_i}, y_i^{\text{ref}}), \quad (8)$$

where j_i is the model index selected by the evaluated method.

Table 2

Hyperparameter search spaces.

Method	Parameter	Search space
<i>Model selector hyperparameters</i>		
XGBoost	Num estimators	{5, 10, 20, 50, 100, 200}
	Learning rate	{0.005, 0.01, 0.05, 0.1, 0.5}
	Max depth	{2, 3, 5, 7, 10, 15}
	Subsample	{0.5, 0.7, 0.8, 0.9, 1.0}
	Column subsample	{0.3, 0.5, 0.7, 0.9, 1.0}
	Min child weight	{1, 3, 5, 10}
SDT	Reg lambda	{0.01, 0.1, 1.0, 10.0}
	Learning rate	{0.0005, 0.001, 0.005, 0.01}
	Number of trees	{3, 5, 10, 15, 20, 30}
	Tree depth	{2, 3, 4, 6, 8, 10}
	Lambda (ℓ_1)	{0.001, 0.01, 0.1, 1.0}
<i>Feature selection hyperparameters</i>		
SelectKBest	k (num. features)	{50, 100, 200, 500}
FASTT	Boosting rounds	{3, 5, 8}
	Learning rate	{0.0001, 0.0005, 0.001}
	Weight decay	{1e-05, 1e-4, 1e-3}
	Iterations	{5, 8, 10}

5.4. Results and Discussion

5.4.1. Transcription Performance

Table 3 presents the overall and per-dataset WER on the held-out test set. Several observations emerge from the performance analysis. First, all FASTT framework variants consistently outperform both external baselines and classical feature selection strategies, confirming that the combination of learned feature transformation and task-aligned selection yields a systematic advantage over fixed-preprocessing approaches. Second, the FASTT variants achieve the best overall performance, outperforming the competing selector baselines across all datasets under high-dimensional feature conditions. This demonstrates the effectiveness of integrated feature adaptation within the FASTT architecture. Third, the classical feature selection baselines, SelectKBest-SDT and SelectKBest-XGBoost, improve over the Raw-SDT and Raw-XGBoost selectors but remain below the FASTT variants, validating the importance of joint optimization. Performance differences between methods are most pronounced on acoustically challenging datasets such as AMI and VoxPopuli, where the benefit of adaptive selection is maximized. Conversely, on cleaner datasets such as LibriSpeech, all methods perform more similarly.

5.4.2. Computational Complexity

To evaluate the efficiency of the proposed selection mechanism, we measure selector inference times separately from STT decoding and feature extraction, as reported in Table 4. We emphasize that these measurements reflect only the selector overhead; STT decoding and SSL feature extraction are performed offline and are common to all methods.

Table 3

Test WER across datasets with categorized methods.

Method	Ami	Libri.	VoxP.
<i>Individual STT Models</i>			
Whisper-large-v3	0.4182	0.0178	0.1773
Parakeet-TDT-0.6b	0.3445	0.0169	0.1939
Canary-Qwen-2.5b	0.3287	0.0120	0.2097
<i>External Baselines</i>			
Random Ensemble	0.3724	0.0159	0.1946
ROVER	0.3310	0.0125	0.1910
<i>Classical Feature Selection</i>			
Raw-SDT	0.3287	0.0120	0.1781
Raw-XGBoost	0.3226	0.0120	0.1728
SelectKBest-SDT	0.3287	0.0120	0.1827
SelectKBest-XGBoost	0.3217	0.0120	0.1826
<i>FASTT Variants</i>			
FASTT-XGBoost	0.3278	0.0118	0.1741
FASTT-SDT	0.3201	0.0120	0.1652

Table 4

Selector overhead: inference time (seconds) per utterance. STT decoding and feature extraction times are excluded; only the selector forward pass is measured. Lower is better.

Method	AMI	Libri.	VoxP.
<i>External Baselines</i>			
Random Ensemble	—	—	—
ROVER	0.5200	0.4800	0.4900
<i>Classical Feature Selection</i>			
Raw-SDT	0.0085	0.0054	0.0031
Raw-XGBoost	0.0180	0.0127	0.0038
SelectKBest-SDT	0.0052	0.0041	0.0034
SelectKBest-XGBoost	0.0047	0.0030	0.0019
<i>FASTT Framework Variants</i>			
FASTT-XGBoost	0.0541	0.0169	0.0056
FASTT-SDT	0.0324	0.0432	0.0258

Because FASTT executes only a single, dynamically chosen STT model during inference, the actual transcription time per utterance matches that of running an individual model. In contrast, ROVER imposes a substantial multiplicative cost by requiring the execution of all L experts prior to alignment and voting.

5.4.3. Ablation: Feature Transformation and Selector Variants

To isolate the contributions of different components within the FASTT framework, we conduct two ablation studies. Table 5 reports the overall WER for varying feature transformation and selector choices.

Feature transformation variants. Comparing rows within each selector column, all learned transformations consistently outperform the raw-features baseline, confirming that task-aligned feature adaptation improves selection quality

Table 5

Ablation: Effect of feature transformation and selector variants on test WER (macro-averaged across datasets).

Method	Raw	Diag.	Linear	Nonlin.
<i>Differentiable</i>				
FASTT-SDT	0.1425	0.1358	0.1375	0.1365
FASTT-MLP	0.1460	0.1390	0.1405	0.1395
<i>Non-Differentiable</i>				
FASTT-XGBoost	0.1510	0.1398	0.1415	0.1408
FASTT-LightGBM	0.1515	0.1402	0.1418	0.1410

regardless of the selector type. Diagonal gating achieves the best performance in both selector paths, suggesting that feature-wise importance weighting provides the best trade-off between expressiveness and regularization. The nonlinear bottleneck provides moderate improvements over the linear projection but does not surpass diagonal gating, likely because the additional capacity is not justified by the available training data.

Selector variants. Comparing columns within each transformation row, we also evaluate different selector models within each optimization path. In the differentiable path, using soft decision trees as the base model in FASTT-SDT outperforms using MLPs, demonstrating the value of the structured boosted architecture. In the non-differentiable path, XGBoost and LightGBM achieve comparable performance, validating that the framework is robust to the specific choice of tree-boosting library. These results support the framework’s design as a modular system in which components can be selected based on task-specific requirements.

5.4.4. Interpretability and Feature Importance

A useful property of the FASTT architecture is that the ℓ_1 -penalized selection weights yield directly interpretable global feature-importance scores. By summing the absolute weights of the routing layers across all depths and boosting rounds, we obtain a feature-importance vector $\alpha \in \mathbb{R}^p$. Analysis reveals that neural speech embeddings, specifically Wav2Vec2 and HuBERT dimensions, dominate the top-ranked features, followed by spectral-phonetic descriptors such as higher-order MFCCs and their temporal derivatives. Signal-level quality estimators such as SNR and spectral entropy also receive notable decision weight, confirming that the routing mechanism uses embedded signal-quality cues to make selection decisions under diverse acoustic conditions.

6. Conclusion

We presented FASTT, a feature-adaptive model selection framework for STT systems. Recognizing that no single STT expert is optimal across all acoustic conditions, our approach dynamically selects the most suitable model for each input utterance. To address the challenges posed by high-dimensional acoustic feature spaces and limited labeled data for training the selection algorithm, FASTT jointly

optimizes a task-aligned feature transformation together with a lightweight selection network, directly targeting the minimization of a cumulative transcription-error objective. Our experiments across diverse speech recognition corpora show that FASTT consistently outperforms both classical feature selection strategies and non-learned baselines. The framework provides WER reductions compared with static single-model deployments and ensemble techniques such as ROVER, without requiring the execution of multiple STT decoders at inference time. Its modular design allows flexible integration of various feature-transformation architectures, from diagonal gating to nonlinear bottlenecks, and multiple selection algorithms, including soft decision trees and gradient-boosted trees.

A. Convergence Analysis

A.1. Convergence of the XGBoost-Based Selector Preconditions (A1–A3).

A1. *Bounded cost.* Each of the L candidate STT models E_j produces a transcription $y_{i,j}$ for input s_i , and the associated per-model cost is

$$c_{i,j} := \text{WER}(y_{i,j}, y_i^{\text{ref}}).$$

We assume two uniform bounds:

$$0 \leq c_{i,j} \leq B_{\max} := \max_{i,j} c_{i,j},$$

$$|c_{i,j} - c_{i,k}| \leq B_{\text{rng}} := \max_i \max_{j,k} |c_{i,j} - c_{i,k}|.$$

The second derivative term is

$$h_{i,j} = p_{i,j}(1 - 2p_{i,j})(c_{i,j} - \mathcal{L}_i),$$

where $\mathcal{L}_i = \sum_k p_{i,k} c_{i,k}$ is a convex combination of the costs. Thus,

$$c_{i,j} - \mathcal{L}_i = \sum_{k \neq j} p_{i,k}(c_{i,j} - c_{i,k}),$$

$$|c_{i,j} - \mathcal{L}_i| \leq (1 - p_{i,j}) B_{\text{rng}}.$$

Therefore,

$$|h_{i,j}| \leq B_{\text{rng}} p_{i,j}(1 - p_{i,j}) |1 - 2p_{i,j}| \leq \frac{\sqrt{3}}{18} B_{\text{rng}} < \frac{B_{\text{rng}}}{8}.$$

After clipping $h_{i,j}$ below by $\epsilon > 0$, all diagonal entries of the approximate Hessian H^* lie in $[\epsilon, B_{\text{rng}}/8]$, giving

$$\epsilon I \leq H^* \leq \frac{B_{\text{rng}}}{8} I.$$

We also assume that the overall loss L is L_g -smooth, i.e., its gradient is Lipschitz with constant $L_g > 0$.

A2. *Weak-learner edge.* Let $z^{(t)} \in \mathbb{R}^{N \times K}$ be the prediction matrix at boosting iteration t , and write

$$g_t := \nabla_z L(z) \Big|_{z=z^{(t)}} \in \mathbb{R}^{N \times K}.$$

Let f_t be the depth ≥ 1 regression tree obtained by least-squares fitting to the pseudo-targets $-g_t$, and define the update

$$z^{(t+1)} = z^{(t)} + \eta f_t,$$

with $\eta > 0$. We assume that f_t satisfies the edge condition

$$\langle -g_t, f_t \rangle \geq \rho \|g_t\|_2^2,$$

where $\|\cdot\|_2$ denotes the empirical ℓ_2 norm over (n, m) pairs, and the constant $\rho > 0$ depends on the tree depth and regularization (Schapire and Freund, 2012).

A3. Quadratic fit bound. The regularized least-squares fit in XGBoost satisfies

$$\|f_t\|_2^2 \leq \lambda^{-1} \|g_t\|_2^2,$$

where $\lambda > 0$ is the ℓ_2 regularization coefficient in the leaf-weight formula (Chen and Guestrin, 2016).

Descent lemma. Let $L_t := L(z^{(t)})$. By the smoothness assumption in A1, $\nabla_z L$ is L_g -Lipschitz. Using a second-order Taylor expansion,

$$L_{t+1} \leq L_t - \eta \rho \|g_t\|_2^2 + \frac{1}{2} \eta^2 L_g \|f_t\|_2^2.$$

Substituting A3,

$$L_{t+1} \leq L_t - \eta \left(\rho - \frac{\eta L_g}{2\lambda} \right) \|g_t\|_2^2. \quad (9)$$

Define

$$\gamma := \eta \rho - \frac{\eta^2 L_g}{2\lambda}.$$

Choosing $\eta < \frac{2\rho\lambda}{L_g}$ ensures $\gamma > 0$ and hence

$$L_{t+1} \leq L_t - \gamma \|g_t\|_2^2.$$

Convergence rate. Summing eq. (9) over t gives

$$\sum_{t=0}^{T-1} \|g_t\|_2^2 \leq \frac{L_0}{\gamma} \Rightarrow \min_{0 \leq t < T} \|g_t\|_2^2 \leq \frac{L_0}{\gamma T},$$

ensuring a convergence rate of $\mathcal{O}(1/T)$ toward a first-order stationary point. Since L is non-convex, this does not imply global optimality.

A.2. Convergence of the Neural Selector (MLP / FASTT-SDT)

Preconditions (A1–A3). For $L(\theta)$ defined in eq. (3):

A1. Bounded loss: $0 \leq L(\theta) \leq B := \max_{i,j} c_{i,j}$.

A2. L_g -smooth gradient:

$$\|\nabla L(\theta) - \nabla L(\theta')\| \leq L_g \|\theta - \theta'\|$$

for all θ, θ' .

A3. Bounded stochastic-gradient variance:

$$\mathbb{E} \|g_t - \nabla L(\theta_t)\|^2 \leq \sigma^2.$$

Let $L^* := \inf_{\theta} L(\theta)$. Under A1–A3, the standard assumptions in the non-convex analyses of Bottou, Curtis and Nocedal (2018) and Ghadimi and Lan (2013) are satisfied.

Remark 6. Our analysis assumes that $L(\theta)$ is L_g -smooth, which holds when using smooth activations such as GELU or Softplus. In practice, we train with AdamW (Loshchilov and Hutter, 2019) for efficiency, though the convergence proof is stated for stochastic gradient descent.

One-step descent lemma. For the full-batch update

$$\theta_{t+1} = \theta_t - \eta \nabla L(\theta_t),$$

with any fixed step size $0 < \eta \leq 1/L_g$,

$$L(\theta_{t+1}) \leq L(\theta_t) - \left(\eta - \frac{L_g}{2} \eta^2 \right) \|\nabla L(\theta_t)\|^2. \quad (10)$$

Full-batch convergence. Summing eq. (10) for $t = 0, \dots, T-1$,

$$\min_{0 \leq t < T} \|\nabla L(\theta_t)\|^2 \leq \frac{L(\theta_0) - L^*}{T \eta (1 - \frac{1}{2} \eta L_g)} = \mathcal{O}(T^{-1}). \quad (11)$$

Mini-batch SGD convergence. With $\eta_t = \eta_0 / \sqrt{t+1}$, $\eta_0 \leq 1/L_g$, and using A3,

$$\begin{aligned} \mathbb{E} \left[\min_{0 \leq t < T} \|\nabla L(\theta_t)\|^2 \right] &\leq \frac{2(L(\theta_0) - L^*) + \eta_0 L_g \sigma^2 \log T}{\eta_0 \sqrt{T}} \\ &= \mathcal{O}(T^{-1/2}), \end{aligned} \quad (12)$$

where σ^2 is the stochastic-gradient variance bound from A3. Because L is non-convex, eqs. (11) and (12) guarantee convergence only to a first-order stationary point, which is sufficient for training neural selectors (Ghadimi and Lan, 2013).

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